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Book Chapters:

St-Onge, B.; Leboeuf, A.; Gougeon, F.A.; Létourneau, J.-P. 2009. [Chapitre 9 - Télédétection](#). Pages 451-479 in Manuel de foresterie. Ordre des ingénieurs forestiers du Québec. Éditions MultiMondes. Québec, Canada.

Gougeon, F.A. 1996. [Chapitre 10 - Télédétection numérique](#). Pages 539-569 in Bérard, J. (ed.); Manuel de foresterie. Ordre des ingénieurs forestiers du Québec. Les Presses de l'Université Laval. Sainte-Foy, Québec, Canada.

Journal Papers:

Wulder, M.A., T. Hermosilla, G. Stinson, F.A. Gougeon, J.C. White, D.A. Hill, B.P. Smiley. 2020. [Satellite-based time series land cover and change information to map forest area consistent with national and international reporting requirements](#). Forestry: An International Journal of Forest Research. Volume 93, Issue 3, April 2020, Pages 331–343. DOI: <http://doi.org/10.1093/forestry/cpaa006> [Open Access]

Trofymow, J.A., Kelley, J., and Gougeon, F. 2019. [Comparison of geospatial and ground-based methods for determining postharvest dispersed woody residues](#). Can. J. For. Res. 49: 1277–1288 [dx.doi.org/10.1139/cjfr-2018-0378](https://doi.org/10.1139/cjfr-2018-0378)

Leckie, D.G., Gougeon F.A., McQueen R., Oddleifson K., Hughes N., Walsworth N., and Gray S. 2017 [Production of a Large-Area Individual Tree Species Map for Forest Inventory in a Complex Forest Setting and Lessons Learned](#) Canadian Journal of Remote Sensing/ Journal canadien de Télédétection DOI: [10.1080/07038992.2017.1286974](https://doi.org/10.1080/07038992.2017.1286974)

Leckie, D.G., Walworth, N.A., and Gougeon, F.A. 2016a. [Identifying Tree Crown Delineation Shapes and Need for Remediation on High Resolution Imagery Using an Evidence Based Approach](#). ISPRS Journal Photogrammetry and Remote Sensing. 114:206:227

Leckie, D.G., N. Walsworth, and F.A. Gougeon. 2016b. [Recognition and Possible Remediation of Automated Tree Delineations with Multiple Isolations per Tree \(Split Cases\) on High Resolution Imagery](#). Canadian Journal of Remote Sensing, 42:1–24, 2016

Leckie, D.G., Walworth, N.A., Gougeon, F.A., Gray, S., Johnson, D., Johnson, L., Oddleifson, K., Plotsky, D, Rogers, V. 2016c. [Automated Individual Tree Isolation on High Resolution Imagery: Possible Methods for Breaking Isolations Involving Multiple Trees](#). IEEE Journal Selected Topics in Applied Earth Observations and Remote Sensing. vol. 9, issue 7, Pages 3229-3248.

Mora, B., M.A. Wulder, G. Hobart, J.C. White, C.W. Bater, F.A. Gougeon, A. Varhola, N.C. Coops. 2013. [Forest inventory stand height estimates from very high spatial resolution satellite imagery calibrated with LIDAR-plots](#).

- Katoh, M.; Gougeon, F.A. 2012. [Improving the precision of tree counting by combining tree detection with crown delineation and classification on homogeneity guided smoothed high resolution \(50 cm\) multispectral aerial digital data.](#) Remote Sensing. 4:1411-1424.
- Ferster, C.J.; Trofymow, J.A.; Coops, N.; Chen, B.; Black, T.A.; Gougeon, F.A. 2011. [Determination of ecosystem carbon-stock distributions in the flux-footprint of an eddy-covariance tower in a coastal forest in British Columbia.](#) Canadian Journal of Forest Research. 41:1380-1393.
- Larsen, M.; Eriksson, M.; Descombes, X.; Perrin, G.; Brandtberg, T.; Gougeon, F.A. 2011. [Comparison of six individual tree crown detection algorithms evaluated under varying forest conditions.](#) International Journal Remote Sensing. 32(20):5827-5852.
- Gougeon, F.A. 2010. [Forest Remote Sensing in Canada and the Individual Tree Crown \(ITC\) Approach to Forest Inventories.](#) Journal of the Faculty of Agriculture, Shinshu Univ., Minamiminowa, Nagano, Japan. 46(1-2):85-92.
- Katoh, M.; Gougeon, F.A.; Leckie, D.G. 2009. [Application of high resolution airborne data using individual tree crowns in Japanese conifer plantations.](#) Journal of Forest Research (Japan). 14:10-19.
- Hyypä, J.; Hyypä, H.; Leckie, D.; Gougeon, F.; Yu, X.; Maltamo, M. 2008. [Review of methods of small-footprint airborne laser scanning for extracting forest inventory data in boreal forests.](#) International Journal Remote Sensing. 29(5):1339-1366.
- Getzin, S.; Wiegand, K.; Schumacher, J.; Gougeon, F.A. 2008. [Scale-dependent competition at the stand level assessed from crown areas.](#) Forest Ecology and Management. 255(7):2478-2485.
- Gougeon, F.A. and Leckie, D.G. 2006 [The individual tree crown approach applied to IKONOS images of a coniferous plantation area.](#) Photogrammetric Engineering and Remote Sensing. 72(11):1287-1297
- Leckie, D.G.; Tinis, S.; Nelson, T.; Burnet, C.N.; Gougeon, F.A.; Cloney E.; Paradine, D. 2005. [Issues in Species Classification of Trees in Old Growth Conifer Stands](#) Can. J. of Remote Sensing. 31(2):175-190.
- Leckie, D.G.; Gougeon, F.A.; Tinis, S.; Nelson, T.; Burnet, C.N.; Paradine, D. 2005. [Automated tree recognition in old growth conifer stands with high resolution digital imagery.](#) Remote Sensing of the Environment. 94:311-326.
- Leckie, D.G.; Jay, C.; Gougeon, F.A.; Sturrock, R.; Paradine, D. 2004. [Detection and assessments of trees with Phellinus weirii \(Laminated root rot\) using high resolution multispectral imagery.](#) Int. J. of Remote Sensing. 25(4):793-818.
- Leckie, D.G.; Gougeon, F.A.; Hill, D.; Quinn, R.; Armstrong, L.; Shreenan, R. 2003. [Combined high density lidar and multispectral imagery for individual tree crown analysis.](#) Can. J. of Remote Sensing. 29(5):1-17.
- Leckie, D.G.; Gougeon, F.A.; Walsworth, N.; Paradine, D. 2003. [Stand delineation and composition estimation using semi-automated individual tree crown analysis.](#) Remote Sensing of Environment. 85:355-369.
- Gougeon, F.A. 1995. [A crown-following approach to the automatic delineation of individual tree crowns in high spatial resolution aerial images.](#) Can. J. Rem. Sens. 21(3):274-284.
- Gougeon, F.A. 1995. [Comparison of possible multispectral classification schemes for tree crowns individually delineated on high spatial resolution MEIS Images.](#) Can. J. Rem. Sens. 21(1):1-9.

Pinz, A.; Zaremba, M.B.; Bischof, H.; Gougeon, F.A.; Locas, M.; 1993. [Neuromorphic methods for recognition of compact image objects](#). Machine Graphics and Vision. 2(3):209-229.

CFS Information Reports:

Gougeon, F.A.; Leckie, D.G; Woods M. 2023. [Individual Tree Crown \(ITC\) Inventory and Analysis of the Petawawa Research Forest: Stand's species composition and ITC-based volume](#). PFC Information Report. BC-X-460. Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C., Canada. 17 p.

Gougeon, F.A.; Labrecque, P.; Guérin, M.; Leckie, D.G. 2019. [Semi-automatic individual tree crown analysis of forests from high spatial resolution satellite images: Comparison with a Quebec ecoforestry map](#). PFC Information Report BC-X-445. Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C., Canada. 24 p.

Trofymow, J.A.; Gougeon, F.A.; Jason Kelley J. 2017. [Determination of Dispersed and Piled Post-harvest Residues in Coastal Douglas-fir Cutblocks Using Unmanned Aerial Vehicle Imagery and Ground-based Surveys](#). CWFC Information Report FI-X-015. Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C., Canada. 27 p.

Gougeon, F.A.; Leckie, D.G. 2003. [Forest information extraction from high spatial resolution images using an individual tree crown approach](#). PFC Information Report. BC-X-396. Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C., Canada. 26 p.

Gougeon, F. 1991. [A forestry expert package: the lake Traverse study](#). PNFI Information Report. PI-X-108. Forestry Canada, Petawawa Natl. For. Inst.; 17 p.

Conference Papers:

Kato M., Gougeon F., Yamamoto S., Nishimura N., Hoshino D., Li Y., 2012 Verification of the accuracy of a new tree species counting method using high resolution satellite data in an old-growth *Chamaecyparis obtusa* forest in central Japan. In ForestSAT 2012, Corvallis (Oregon), USA, Sept.11-14, 2012.

Leckie D., Woods M., Pitt D., Gougeon F., Lacroix S., Luther J., Whitehead R., Groot A., Walsworth N., 2012. Progress towards an enhanced forest inventory and integration into the forest value chain. In 33rd Canadian Symposium on Remote Sensing, Ottawa(Ontario), Canada, June 11-14, 2012.

Gougeon, F.A. and Leckie, D.G. 2011. [ITC Analyses of the Petawawa Research Forest from Satellite and Aerial Data](#) In 32nd Canadian Symposium on Remote Sensing / 14e Congrès de l'Association québécoise de télédétection, Sherbrooke(Québec), Canada, June 13-16, 2011.

Zhang C., Collins M.J., Leckie D.G., Gougeon F.A. 2009. [The Synthetic Crown Recognition Model for Automatic Image Interpretation on UltraCamD Imagery](#). Pages 1159-1165 in 30th Canadian Symposium on Remote Sensing, Lethbridge, Alberta, Canada, June 22-25 2009.

Gougeon, F.A. 2009. [The Individual Tree Crown \(ITC\) Approach to Forest Inventories: Satellite and Aerial Sensor Considerations](#). In IUFRO Div. 4 Conf.: Extending Forest Inventory and Monitoring, Québec City, Canada, May 19-22, 2009.

Gougeon, F.A. 2008. [Les images aériennes de haute résolution spatiale pour l'inventaire forestier semi-automatique](#). In 13ème Congrès de l'Association Québécoise de Télédétection, Trois-Rivières (Québec), Canada, April 30 - May 1,

- Zaremba, M.B. and Gougeon, F.A. 2006. [Kernel-based statistical learning for individual tree recognition in forest inventory](#) GeoTec Event 2006 Conference, Ottawa (Ontario), June 18 - 21, 2006.
- Zaremba, M.B. and Gougeon, F.A. 2006. [Fusion of High-Resolution Satellite and Lidar Data for Individual Tree Recognition](#) IEEE Canadian Conference on Electrical and Computer Engineering, Ottawa (Ontario), May 2006.
- Gougeon, F.; Cormier, R.; Labrecque, P.; Cole, B.; Pitt, D.; Leckie, D. 2003. [Individual Tree Crown \(ITC\) Delineation on Ikonos and QuickBird Imagery: the Cockburn Island Study](#). In 25rd Canadian Symposium on Remote Sensing / 11e Congrès de l'Association québécoise de télédétection, Montréal (Québec), Canada, Oct. 14-17, 2003.
- Leckie, D.G.; Cloney, E.; Gougeon, F.A.; Hill, D.; Shand A.; Alfaro R. 2003. [Spruce Budworm Damage Assessment in Northern Forests from High Resolution Satellite Imagery](#). In 25rd Canadian Symposium on Remote Sensing / 11e Congrès de l'Association québécoise de télédétection, Montréal (Québec), Canada, Oct. 14-17, 2003.
- Gougeon, F.; Labrecque, P.; Guérin, M.; Leckie, A.; Dawson, A. 2001. [Détection du pin blanc dans l'Outaouais à partir d'images satellitaires à haute résolution IKONOS](#). In Proc. 23rd Canadian Symposium on Remote Sensing / 10e Congrès de l'Association québécoise de télédétection (CD-ROM), Sainte-Foy, Québec, Canada, August 21-24, 2001.
- Gougeon, F.A.; St-Onge, B.A.; Wulder, M.; Leckie, D. 2001. [Synergy of Airborne Laser Altimetry and Digital Videography for Individual Tree Crown Delineation](#). In Proc. 23rd Canadian Symposium on Remote Sensing / 10e Congrès de l'Association québécoise de télédétection (CD-ROM), Sainte-Foy, Québec, Canada, August 21-24, 2001.
- Gougeon, F.A.; D. Leckie. 2001. [Individual Tree Crown Image Analysis: A Step Towards Precision Forestry](#). In Proc. of First Int. Precision Forestry Symposium (CD-ROM), Seattle, Washington, USA, June 17-20, 2001.
- Bhogal, A.S.; Goodenough, D.G.; Gougeon, F.; Dyk, A.; Nieman, O. 2000. [AVIRIS imagery for forest attribute information: anisotropic effects and limitations in multitemporal data](#). Pages 1388-1392 in Proc. 2000 Int'l. Geoscience and Remote Sensing Symposium (IGARSS'00). Hawaii. July 24-28, 2000.
- Leckie, D.; Smith, N.; Davison, D.; Jay, C.; Gougeon, F.; Achal, S.; Burnett, C.; Cloney, E.; Lataille, S.; Montgomery, F.; Nelson, T.; Walsworth, N. 1999. [Automated interpretation of high spatial resolution multispectral \(CASI\) imagery: A development project for a forest company](#). Pages 201-211 in Proc. Fourth Int. Airborne Rem. Sens. Conf. and Exh. / 21st Can. Symp. Rem. Sens, Vol. I. Ottawa, Canada, June 21-24, 1999.
- Leckie, D.; Burnett, C.; Nelson, T.; Jay, C.; Walsworth, N.; Gougeon, F.; Cloney, E. 1999. [Forest parameter extraction through computer-based analysis of high resolution imagery](#). Pages 205-213 in Proc. Fourth Int. Airborne Rem. Sens. Conf. and Exh. / 21st Can. Symp. Rem. Sens, Vol. II. Ottawa, Canada, June 21-24, 1999.
- Leckie, D.; Gougeon, F.; Jay, C.; Tam, A.; Leach, R.; Walsworth, N. 1999. [A flexible view angle BRDF compensation system using aircraft orientation information](#). Pages 115 in Proc. Fourth Int. Airborne Rem. Sens. Conf. and Exh. / 21st Can. Symp. Rem. Sens, Vol. I. Ottawa, Canada, June 21-24, 1999.
- Magnussen, S.; Gougeon, F.; Leckie, D.; Wulder, M. 1999. [Predicting tree heights from a combination of LIDAR canopy heights and digital stem counts](#). In IUFRO Conf. on Rem. Sens. and Monitoring, June 1-3, 1999. Rogow, Poland.
- Gougeon, F.A. 1999. [Automatic individual tree crown delineation using a valley-following algorithm and rule-based system](#). Pages 11-23 in Hill, D.A. and Leckie, D.G.; eds. Proc. Int'l Forum on Automated Interpretation of High Spatial

Resolution Digital Imagery for Forestry. February 10-12, 1998. Victoria, B.C.; Canada. Natural Resources Canada, Canadian Forest Service, Victoria, B.C.; Canada.

Gougeon, F.A.; Leckie, D.; Scott, I.; Paradine, D. 1999. [Individual tree crown species recognition: the Nahmint study](#). Pages 209-223 in Hill, D.A. and Leckie, D.G.; eds. Proc. Int'l Forum on Automated Interpretation of High Spatial Resolution Digital Imagery for Forestry. February 10-12, 1998. Victoria, British Columbia, Canada. Natural Resources Canada, Canadian Forest Service, Victoria, B.C.; Canada.

Gougeon, F.A.; D. Leckie. 1999. [Forest regeneration: individual tree crown detection techniques for density and stocking assessment](#). Pages 169-177 in Hill, D.A. and Leckie, D.G.; eds. Proc. Int'l Forum on Automated Interpretation of High Spatial Resolution Digital Imagery for Forestry. February 10-12, 1998. Victoria, British Columbia, Canada. Natural Resources Canada, Canadian Forest Service, Victoria, B.C.; Canada.

Leckie, D.G.; Gillis, M.; Gougeon, F.; Lodin, M.; Wakelin, J.; Yuan, X. 1999. [Computer-assisted photointerpretation aids to forest inventory mapping: some possible approaches](#). Pages 335-343 in Hill, D.A.; Leckie, D.G.; eds. Proc. Int'l Forum on Automated Interpretation of High Spatial Resolution Digital Imagery for Forestry. February 10-12, 1998. Victoria, British Columbia, Canada. Natural Resources Canada, Canadian Forest Service, Victoria, B.C.; Canada.

Leckie D.G.; Gougeon, F.A. 1999. [An assessment of both visual and automated tree counting and species identification with high spatial resolution multispectral imagery](#). Pages 141-152 in Hill, D.A. and Leckie, D.G.; eds. Proc. Int'l Forum on Automated Interpretation of High Spatial Resolution Digital Imagery for Forestry. February 10-12, 1998. Victoria, B.C.; Canada. Natural Resources Canada, Canadian Forest Service, Victoria, B.C.; Canada.

Bhogal, P.; Goodenough, D.G.; Gougeon, F.; Daley, N.; Nera, A.; Nieman, O. 1998. [Fusion of AIRSAR and MEIS data for estimating forest parameters](#). In Proc. 1998 Int'l. Geoscience and Remote Sensing Symposium (IGARSS'98). July 6-10. Seattle, Washington. 4 p.

Gougeon, F.A. 1997. [Recognizing the forest from the trees: individual tree crown delineation, classification and regrouping for inventory purposes](#). Pages 807-814 in Proc. Third Int. Airborne Rem. Sens. Conf. and Exh., Vol. II. Copenhagen, Denmark, July 7-10, 1997.

Gougeon, F.A. 1997. [A locally adaptive technique for forest regeneration assessments from high resolution aerial images](#). In CD-ROM of Proc. 19th Can. Symp. Rem. Sens. Ottawa, Ontario, Canada, May 24-30, 1997. 9 p.

Gougeon, F.A. 1996. [Vers l'inventaire forestier automatisé: reconnaître l'arbre ou la forêt?](#) In CD-ROM of 9e Congrès de L'Association québécoise de télédétection. Québec, Canada, May 1-3 1996. 10 p.

Gougeon, F.A. 1995. [A system for individual tree crown classification of conifer stands at high spatial resolution](#). Pages 635-642 in Proc. 17th Can. Symp. Rem. Sens. Saskatoon, Saskatchewan, Canada, June 13-15, 1995.

Gougeon, F.A.; Kourtz, P.; Strome, M. 1994. [Preliminary research on robotic vision in a regenerating forest environment](#). Pages 255-262 in Borkowski, A.; Crowley, J.L.; eds. Proc. Intl. Symp. Intelligent Robotic Systems '94. Grenoble, France, July 11-15, 1994.

Gougeon, F.A. 1993. [Individual tree identification from high resolution MEIS images](#). Pages 117-128 in Leckie, D.G.; Gillis, M.D.; eds. Intl. Forum Airborne Multispectral Scanning for Forestry and Mapping (with Emphasis on MEIS). Val-Morin, Québec, Canada, April 13-16, 1992. Inf. Rep. PI-X-113, Forestry Canada, Petawawa Natl. For. Inst.

Zaremba, M.B.; Locas, M.; Gougeon, F.A. 1993. [La reconnaissance des arbres sur une image multispectrale de haute résolution par des réseaux neuronaux](#). Pages 757-760 in Proc. 16th Can. Symp. Rem. Sens. / 8e congrès de

L'association québécoise de télédétection. Sherbrooke, Québec, Canada, June 7-10, 1993.

Kourtz, P.; Strome, M.; Gougeon, F. 1993. Autonomous forest stand-tending robots. Pages 70-78 in Köhl, M.; Gertner, G.Z.; eds. Proc. Mtng Stat. Methods, Mathematics and Computers. IUFRO S4.11-00 Centennial Congress. Berlin/Eberswalde, Germany, WSL/FNP, Birmensdorf, Switzerland, Aug. 31- Sept. 6, 1992.

Strome, M.; Kourtz, P.; Gougeon, F. 1992. [Robotics for silviculture operations](#). Pages 236-241 in Proc. Forest Sector 2000 - The Application of Advanced Informatics to the Forest Industry. Ind.; Sci. Tech. Can.; Vancouver, Canada, May 25-27, 1992.

Kourtz, P.; Strome, M.; Gougeon, F. 1992. Autonomous forestry stand-tending robots. In Proc. Canada-France Workshop in Intelligent Robotics. Ottawa, Ontario, Canada, May 19-21, 1992. 6 p.

Gougeon, F.A. 1991. [Le suivi forestier avec le Progiciel expert en foresterie](#). Pages 153-162 in Gagnon, P. ed. Proc. 7e congrès de L'association québécoise de télédétection. Montréal, Québec, Canada, 23-25 octobre 1991.

Gougeon, F.A.; Moore, T. 1989. [Classification individuelle des arbres à partir d'images à haute résolution spatiale](#). Pages 185-196 in Bernier, M.; et al.; eds. Télédétection et gestion des ressources Vol. VI - 6e congrès de L'association québécoise de télédétection. Sherbrooke, Québec, Canada, May 4-6, 1988.

Gougeon, F.A.; Moore, T. 1988. [Individual tree classification using MEIS-II imagery](#). Page 927 in Proc. 1988 Intl. Geoscience and Remote Sensing Symposium (IGARSS'88). Edinburgh, Scotland, September 13-16, 1988.

Gougeon, F.A.; Wong, A.K.C. 1987. [Towards knowledge-based spectral and textural signature recognition](#). Pages 565-578 in P. Howarth, ed. Proc. 11th Can. Symp. Rem. Sens. Waterloo, Ontario, Canada, June 22-25, 1987.

Gougeon, F.A.; Wong, A.K.C. 1986. [Spectral and textural segmentation of multispectral aerial images](#). Pages 291-300 in Thompson, D.; Brown, R.J.; eds. Proc. 10th Can. Symp. Rem. Sens. Edmonton, Alberta, Canada, May 5-8, 1986.

Leckie, D.G.; Gougeon, F.A. 1981. [Assessment of spruce budworm defoliation using digital airborne MSS data](#). Pages 190-196 in Best, W.G.; Weselake, S.-A.; ass. eds. Proc. 7th Can. Symp. Rem. Sens. Winnipeg, Manitoba, Canada, Sept. 8-11, 1981.

Leckie, D.G.; Gougeon, F.A.; Kourtz, P.H. 1981. Preliminary results of a study to determine the potential of Landsat-D for forest type mapping and spruce budworm defoliation assessment. Page A47 in Proc. Colloqu. Application of the Next Generation of Earth Resource Satellites. Montréal, Québec, Canada, March 1981.

Goldberg M.; Gougeon, F. 1980. Digital image processing hardware. Pages 417-420 in Proc. IEEE Canadian Communications and Power Conf. Montréal, Québec, Canada, Oct. 15-17, 1980.

Gougeon, F.; Goldberg M. 1979. The multimicroprocessor array computing structure for image processing in Proc. 9th Intl. Symp. Mini and Microcomputers (MIMI 79). Montréal, Québec, Canada, September 24-26, 1979.

Technical Reports:

Gougeon, F.A. 2010. [The ITC Suite Manual : A Semi-Automatic Individual Tree Crown \(ITC\) Approach to Forest Inventories](#). Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C. Canada. June 2010. 92 p.

Chubey, M.; Stehle, K.; Albricht, R.; Gougeon, F.; Leckie, D.; Gray, S.; Woods, M.; Courville, P. 2009. [Semi-Automated Species Classification in Ontario Great Lakes - St. Lawrence Forest Conditions. Final Report: Great Lakes - St. Lawrence ITC Project\(2005/2008\).](#) Ontario Ministry of Natural Resources. January 2009. 71 p.

Gougeon, F. 2005. The Individual Tree Crown (ITC) Suite Manual. Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C. Canada. June 2005. 50 p.

Alfaro, R.; Hill, D.; Hodgkinson, R.; Gougeon, F.; Leckie, D.; Magnussen, S.; Shand, A.; Taylor, S. 2003. Spruce budworm disturbance of the boreal forests in the Ft. Nelson District of British Columbia: Assessing forest damage, risk and ecosystem changes. Report to Forest Innovation and Investment Forest Research Program (RO-32) Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C.; Canada.. April 2003.

Leckie, D.; Gougeon, F.; Hill, D.; Jordan, R.; Morrison, A. 2002. Automated stand delineation for forest inventory mapping in British Columbia: with IKONOS imagery and digital orthophoto mosaics. Final report for BC Ministry of Forests and Ministry of Sustainable Resource Management. Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C.; Canada. June 2002. 22 p.

Alfaro, R.; Leckie, D.; Gougeon, F.; Magnussen, S.; Shand, A.; Taylor, S. 2002. Spruce budworm disturbance of the boreal forests in the Ft. Nelson District of British Columbia: Assessing forest damage, risk and ecosystem changes. Forest Renewal BC Number: PAR02003-16 Report. Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C.; Canada. March 2002. 20 p.

Labrecque, P.; Gougeon, F.; Guérin, M.; Mouton, J.-F. 2002 Vers une meilleure connaissance forestière grâce aux images satellitaires à haute résolution IKONOS - Phase II. Rapport final - Programme de mise en valeur des ressources du milieu forestier Volet 1. CLC-Camint, Hull. Avril 2002. 88 p.

Leckie, D.; C. Jay, D. Hill, A. Shand, F. Gougeon, J. Beaubien; R. Alfaro. 2001. Remote Sensing of Eastern Spruce Budworm in the Fort Nelson Region. Report to B.C. Ministry of Forest (Contract #FR01RPG-024). Pacific Forestry Centre. Natural Resources Canada, Canadian Forest Service, Victoria. B.C.; Canada. May 2001

Labrecque, P.; Gougeon, F.; Guérin, M. 2001. Application d'une méthode d'identification géospatiale du pin blanc à partir d'images satellitaires à haute résolution IKONOS. Rapport final - Programme de mise en valeur des ressources du milieu forestier Volet 1. CLC-Camint, Hull. Avril 2001. 34 p.

Dawson A.; F. Gougeon. 2000. Procedure for Individual Tree Crown (ITC) Analysis of an Image. Pacific Forestry Centre. Natural Resources Canada, Canadian Forest Service, Victoria. B.C.; Canada. December 2000. 25 p.

Gougeon, F.A. 1997. Robotic vision in a regenerating forest environment. Natural Resources Canada, Canadian Forest Service, Pacific Forestry Centre, Victoria, B.C. Canada. July 1997. 26 p.

Zaremba, M.B.; Locas, M.; Monzair M.; Gougeon, F.A. 1993. La reconnaissance des espèces de conifères sur une image MEIS haute définition par les réseaux neuronaux. Research Report RR93/03-4. Dépt. d'informatique. Univ. du Québec à Hull. March 1993. 14 p.

Pinz, A.; Zaremba, M.B.; Bischof, H.; Gougeon, F.A.; Locas, M.; 1993. Neuromorphic methods for recognition of compact image objects. Research Report PRIP-TR-21. Dept. Pattern Recognition and Image Processing, Inst. for Automation, Technical Univ. Vienna.; Vienna, Austria, March 15, 1993. 18 p.

Gougeon, F.A. 1985. A knowledge-based automatic multispectral images information extraction system. File Report. University of Waterloo, Dept. Systems Design Engineering, April 1985, 22 p.

Gougeon, F.A. 1984. Spectral signature extension for Landsat data classification. File Report. University of Waterloo, Dept. Systems Design Engineering, December 1984, 15 p.

Gougeon, F.A. 1984. Probabilistic reasoning in inference systems: the Theory of Evidence. File Report. University of Waterloo, Dept. Systems Design Engineering, August 1984, 21 p.

Gougeon, F.A. 1979. MACS simulator user manual. MACS project report. No. 4, University of Ottawa, Dept. Electrical Engineering, March 31, 1979, 51 p.

Gougeon, F.A. 1978. MACS-1 system simulator specification. MACS project report. No. 1. University of Ottawa, Dept. Electrical Engineering, October 1978, 26 p.

Theses:

Gougeon, F.A. 1987. Texture-based multispectral image segmentation and signature recognition. Ph.D. Thesis. University of Waterloo, Dept. Systems Design Engineering, 128 p.

Gougeon, F.A. 1979. Image processing by parallel processors. M.A.Sc. Thesis. University of Ottawa, Dept. Electrical Engineering, 192 p.

Other Publications:

Trofymow, J.A.; Gougeon F.A.; Kelley J. 2017. [Carbon Transfers and Emissions Following Harvest and Pile Burning in Coastal Douglas-fir Forests Determined from Analysis of High-Resolution UAV Imagery and Point Clouds and from Field Surveys](#). Poster at American Geophysical Union(AGU) 2017 Conference, Dec. 11-15, 2017, New Orleans, USA.

Kurowski, M; Gougeon, F.; 2017. Evaluating Regeneration of Softwood Forest along Resource Roads. Information Note. FPIinnovations. 2 p.

Trofymow, J.A.; Gougeon F.A.; Kelley J. 2015. [Dispersed and piled woody residues volumes in coastal Douglas-fir cutblocks determined using high-resolution imagery from a UAV and from ground-based surveys](#). Poster at American Geophysical Union(AGU) 2015 Conference, Dec. 14-18, 2015, San Francisco, USA.

Gougeon, F. 2013. [Individual tree crown \(ITC\) techniques](#). World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service, Ottawa.

Gougeon, F. 2013. [ITC analysis of satellite images](#). World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service, Ottawa.

Gougeon, F. 2013. [ITC analysis of aerial images](#). World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service, Ottawa.

Gougeon, F. 2013. [Forest regeneration assessment techniques](#). World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service, Ottawa.

Gougeon, F. 2012. [An individual tree approach to forest inventory](#). World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service, Ottawa.

Gougeon, F. 2011. Individual Tree Crown Analysis of High Resolution Satellite Images. World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service. Pacific Forestry Centre, Victoria. 14 p.

Gougeon, F. 2011. Individual Tree Crown Analysis of Aerial Images World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service. Pacific Forestry Centre, Victoria. 10 p.

Gougeon, F. 2000. [Towards semi-automated forest inventories using individual tree crown \(ITC\) recognition](#). Technology Transfer Note - Forestry Research Applications. Pacific Forestry Centre. Natural Resources Canada, Canadian Forest Service, Victoria. B.C.; Canada. No. 22. April, 2000. 6 p.

Gougeon, F. 1999. A software suite for individual tree crown (ITC) recognition and forest analysis. PFC Fact Sheet, Canadian Forest Service, Pacific Forestry Centre. Victoria. 1 p.

Gougeon, F.A. 1997. Individual Tree Crown (ITC) Delineation, Identification and Regrouping in High Spatial Resolution (30-100 cm/pixel) Aerial and Satellite Multispectral Images. World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service. Pacific Forestry Centre, Victoria. 12 p.

Gougeon, F.A. 1997. Forest Regeneration Assessment Techniques. World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service. Pacific Forestry Centre, Victoria. 5 p.

Gougeon, F.A. 1996. Individual Tree Crown Delineation and Identification in High Spatial Resolution (30-100 cm/pixel) Aerial and Satellite Multispectral Images. World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service. Pacific Forestry Centre, Victoria. 8 p.

Gougeon, F.A. 1995. Towards semi-automatic individual tree based forest inventories. World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service. Petawawa National Forestry Institute, Chalk River. 3 p.

Gougeon, F.A. 1995. Robotic vision in a regenerating forest environment. World Wide Web hypertext document. Natural Resources Canada, Canadian Forest Service. Petawawa National Forestry Institute, Chalk River. 6 p.

Leckie, D.G.; Gougeon, F.A. 1982. Mapping and quantifying spruce budworm defoliation: digital remote sensing classification techniques applied to airborne multispectral scanner data. Pages 1-2 in CANUSA Newsletter, No. 24, September 1982.

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